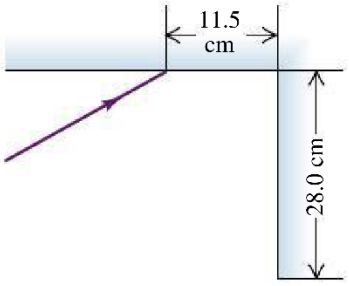
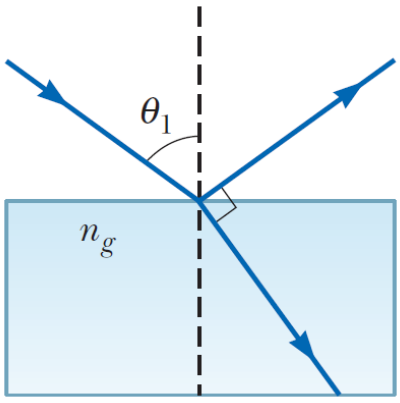
Sheet 1

1. Two plane mirrors intersect at right angles. A laser beam strikes the first of them at a point $11.5\ \text{cm}$ from their point of intersection, as shown in figure. For what angle of incidence at the first mirror will this ray strike the midpoint of the second mirror (which is $28.0\ \text{cm}$ long) after reflecting from the first mirror?



2. A beam of light has a wavelength of $650\ \text{nm}$ in vacuum.
   (a) What is the speed of this light in a liquid whose index of refraction at this wavelength is 1.47?
   (b) What is the wavelength of these waves in the liquid?

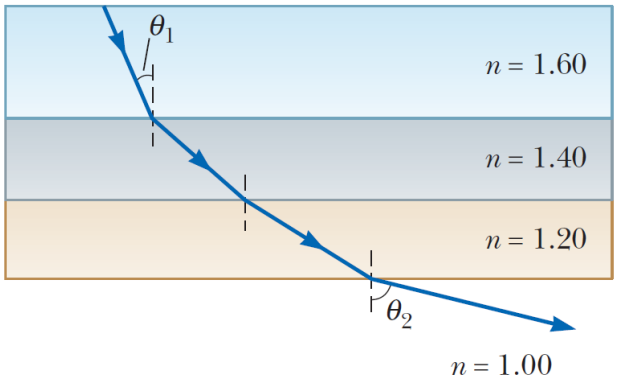
3. A beam of light both reflects and refracts at the surface between air and glass as shown in Figure below. If the refractive index of the glass is n_g, find the angle of incidence θ_1 in the air that would result in the reflected ray and the refracted ray being perpendicular to each other.



4. The figure below shows the path of a light beam through several slabs with different indices of refraction.

   (a) If $\theta_1 = 30.0°$, what is the angle θ_1 of the emerging beam?

   (b) What must the incident angle θ_1 be to have total internal reflection at the surface between the medium with $n = 1.20$ and the medium with $n = 1.00$?



5. The critical angle for total internal reflection at a liquid–air interface is $42.5°$.
   (a) If a ray of light traveling in the liquid has an angle of incidence at the interface of $35.0°$, what angle does the refracted ray in the air make with the normal?
   (b) If a ray of light traveling in air has an angle of incidence at the interface of $35.0°$, what angle does the refracted ray in the liquid make with the normal?

6. The light beam in figure strikes surface 2 at critical angle. Determine the angle of incidence θ_1.

